

Random Variables III: Continuous Distributions & CLT

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Stat 141 | Fall 2026

Wednesday, Week 11

Reminders/Announcements

- Change to my office hours today (when in doubt check Moodle)
- Colloquium tomorrow - statistics PhD student at UW!

Goals for Today

- Review: Binomial distribution activity
- Discuss continuous random variables
- Introduce the normal distribution
- Discuss the **Central Limit Theorem**!

Activity: Survey of Portlanders

Suppose you believe 40% of Portlanders think the city should install more bike lanes. You will take a simple random survey of 50 Portlanders to test your belief.

1. Assume your belief is true. Let X be the number of survey respondents who think the city should install more bike lanes. Does X follow a Binomial distribution? If so, what are n and p?
2. Calculate the expected value and standard deviation of X (still assuming your belief is true)
3. Suppose you conducted the survey and found 30 respondents wanted more bike lanes. In R, how would we calculate the probability that $X \geq 30$? (If you have your computer - test it!)
4. Draw the connection between the probability you calculated and a hypothesis test.

05:00

Continuous Random Variables

The Distribution of a Continuous Variable

If X is a **continuous random variable**, it can take on any value in an **interval**.

- e.g., $0 \leq X \leq 10$ or $-\infty < X < \infty$

Recall: For discrete random variables, we could list the probability of each possible outcome.

- For **continuous** random variables, this won't work. There are an uncountable number of outcomes!

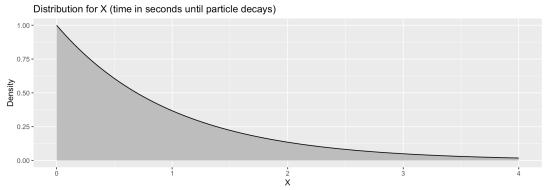
Instead, we represent relative chances of different possible outcomes using a **density** function $f(x)$

- $f(x) \geq 0$ for all possible X
- The total area under the function is 1
- $P(a \leq X \leq b)$ **is the area under** f **between** a **and** b

Density Curve

Suppose X is a random variable representing the time (in seconds) it takes for a particle to experience radioactive decay, where

$$f(x) = e^{-x} \qquad \text{for } x \geq 0$$

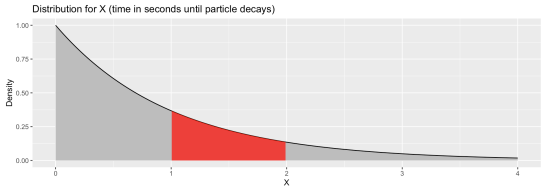


- The probability that it takes between 1 and 2 seconds to decay is the area under the curve between 1 and 2. $P(1 < T < 2) =$

Density Curve

Suppose X is a random variable representing the time (in seconds) it takes for a particle to experience radioactive decay, where

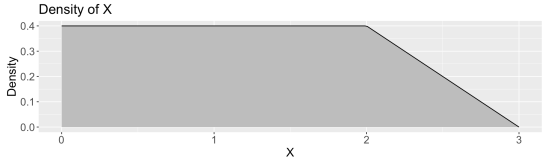
$$f(x) = e^{-x} \qquad \text{for } x \geq 0$$



- The probability that it takes between 1 and 2 seconds to decay is the area under the curve between 1 and 2. $P(1 < T < 2) = 0.232$

Practice with Continuous Random Variables

Let X be a continuous random variable with the density function,

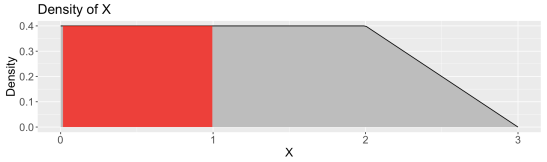


Calculate:

1. $P(X \leq 1)$
2. $P(X \leq 2)$
3. $P(2 \leq X \leq 3)$

Practice with Continuous Random Variables

Let X be a continuous random variable with the density function,

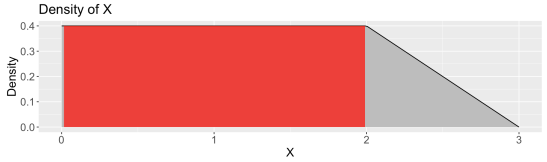


Calculate:

1. $P(X \leq 1) = 1 * 0.4 = \boxed{0.4}$
2. $P(X \leq 2)$
3. $P(2 \leq X \leq 3)$

Practice with Continuous Random Variables

Let X be a continuous random variable with the density function,

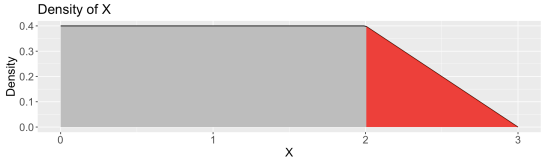


Calculate:

1. $P(X \leq 1) = 1 * 0.4 = \boxed{0.4}$
2. $P(X \leq 2) = 2 * 0.4 = \boxed{0.8}$
3. $P(2 \leq X \leq 3)$

Practice with Continuous Random Variables

Let X be a continuous random variable with the density function,



Calculate:

1. $P(X \leq 1) = 1 * 0.4 = \boxed{0.4}$
2. $P(X \leq 2) = 2 * 0.4 = \boxed{0.8}$
3. $P(2 \leq X \leq 3) = \frac{1}{2} * 1 * 0.4 = \boxed{0.2}$

Mean and Variance

Continuous variables have a **mean, variance, and standard deviation** too!

- We can't use the same definition as for discrete random variables.

$$E[X] = \sum_x xP[X = x]$$

1. There are infinitely many values
2. With infinitely many, each specific value has probability 0 of occurring

Instead, we have to use *calculus* to define mean and variance:

$$E[X] = \int x f(x)\,dx$$
$$\text{Var}(X) = \int (x-\mu)^2 f(x)\,dx$$

Mean and Variance

$$E[X] = \int x f(x)\,dx$$
$$\text{Var}(X) = \int (x-\mu)^2 f(x)\,dx$$
$$\text{SD}(X) = \sqrt{\text{Var}(X)}$$

- These **integrals** are tools to meaningfully average infinitely many values
- *(We won't compute any integrals in this class!)*

As always…

- the **mean** of a random variable represents a **typical value**.
- the **standard deviation** represents the typical size of deviations from the mean.

The Normal Distribution

The Normal Distribution

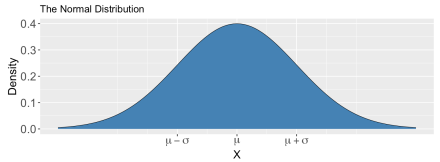
The **Normal distribution** is defined by two parameters:

1. Mean, μ
2. Standard deviation, σ

Suppose X follows a Normal(μ,σ) distribution. The density function is

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \cdot \exp\left(\frac{-(x-\mu)^2}{2\sigma^2}\right)$$

(Don't memorize this!)



This is the official name for "bell curves"!

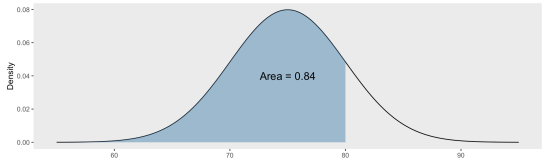
Calculating Probabilities

R has built-in functions for calculating probabilities from a normal distribution.

Suppose $X \sim \text{Normal}(\mu = 75, \sigma = 5)$. Then:

- $P(X < 80) =$

```
1 pnorm(q = 80, mean = 75, sd = 5)
[1] 0.8413447
```



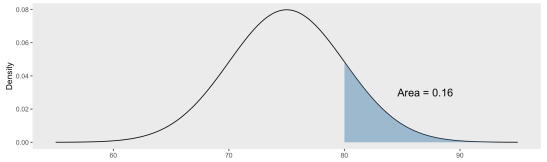
Calculating Probabilities

R has built-in functions for calculating probabilities from a normal distribution.

Suppose $X \sim \text{Normal}(\mu = 75, \sigma = 5)$. Then:

- $P(X \geq 80) = P(X > 80) =$

```
1 1-pnorm(q = 80, mean = 75, sd = 5)
[1] 0.1586553
```



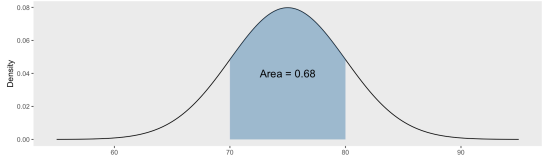
Calculating Probabilities

R has built-in functions for calculating probabilities from a normal distribution.

Suppose $X \sim \text{Normal}(\mu = 75, \sigma = 5)$. Then:

- $P(70 \leq X \leq 80) = P(X \leq 80) - P(X \leq 70) =$

```
1 pnorm(q = 80, mean = 75, sd = 5) - pnorm(q = 70, mean = 75, sd = 5)
[1] 0.6826895
```



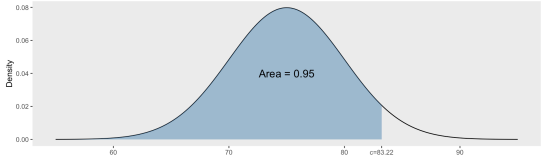
Finding Quantiles

We can also use R to find **quantiles** of a Normal distribution.

Suppose $X \sim \text{Normal}(\mu = 75, \sigma = 5)$. Then:

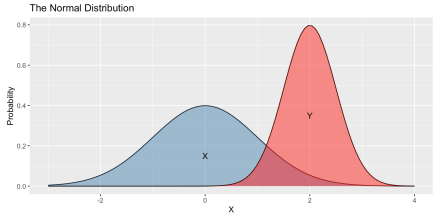
- The 0.95 quantile is the value c such that $P(X < c) = 0.95$

```
1 qnorm(p = 0.95, mean = 75, sd = 5)
[1] 83.22427
```



Scale and Translation Invariance

Suppose $X \sim \text{Normal}(\mu = 0, \sigma = 1)$ and $Y \sim \text{Normal}(\mu = 2, \sigma = 0.25)$.



X and Y have different means, heights, and widths…

- But the **same shapes!**

Scale and Translation Invariance

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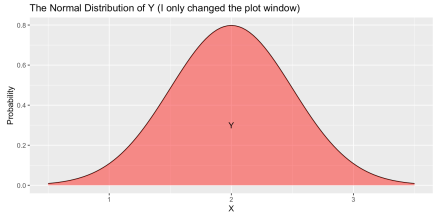


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Scale and Translation Invariance

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X and Y have different means, heights, and widths…

- But the **same shapes!**

Standardization

> **Theorem: Standardization**
> Suppose $X \sim \text{Normal}(\mu, \sigma)$. Then, $Z = \frac{X - \mu}{\sigma}$ is a Normal random variable with mean 0 and standard deviation 1.

Standard Normal: a Normal random variable with mean 0 and standard deviation 1.

- We call the process of subtracting off μ and dividing by σ, **standardizing.**
- Useful Fact: If $X \sim \text{Normal}(\mu = 100, \sigma = 10)$ and $Z \sim \text{Normal}(\mu = 0, \sigma = 1)$, then,

$$P\Big[X < 90\Big] = P\Big[X < \text{ 1 SD below } \mu\Big] = P\Big[Z < -1\Big]$$

More Generally: If $X \sim \text{Normal}(\mu, \sigma)$ and $Z \sim \text{Normal}(0, 1)$, then

$$P(X \leq x) = P\left(Z \leq \frac{x - \mu}{\sigma}\right)$$

Practice with Standard Normals

More Generally: If $X \sim \text{Normal}(\mu, \sigma)$ and $Z \sim \text{Normal}(0, 1)$, then

$$P(X \leq x) = P\left(Z \leq \frac{x - \mu}{\sigma}\right)$$

Practice: For each of the following, rewrite $P(X \leq x)$ as $P(Z \leq z)$ for some z.

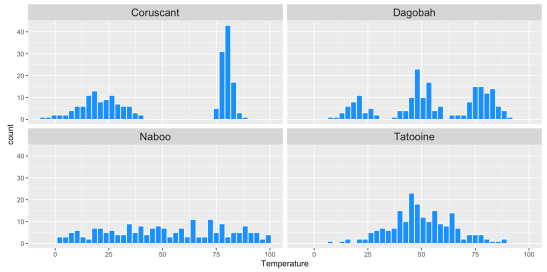
1. $P(X \leq 10)$ where $X \sim \text{Normal}(\mu = 5, \sigma = 5)$
2. $P(X \leq 15)$ where $X \sim \text{Normal}(\mu = 20, \sigma = 10)$

02:00

The Central Limit Theorem

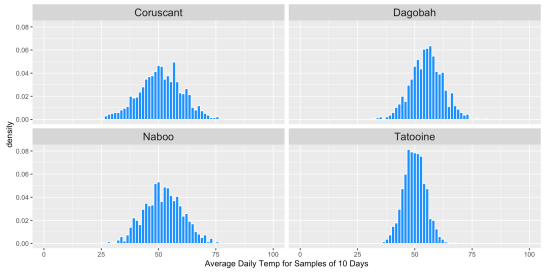
Daily Temperatures

Suppose we're astronomers studying four far away planets: Naboo, Tatooine, Coruscant, and Dagobah. We have the daily temperature on each of these planets over 200 days:



Random Sample Means (n=10)

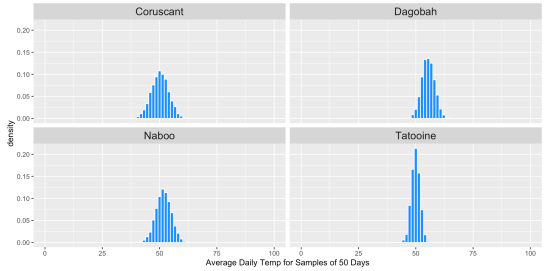
Suppose we repeatedly take samples of **10 days** from each planet, and compute the average temperature $\bar{x}$ for each sample (these are **sampling distributions**):



- **Q**: What does the distribution of sample means look like?

Random Sample Means (n=50)

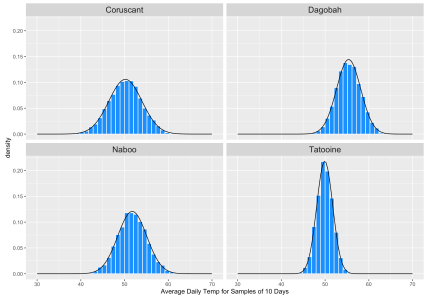
Suppose we repeatedly take samples of **50 days** from each planet, and compute the average temperature $\bar{x}$ for each sample (these are **sampling distributions**):

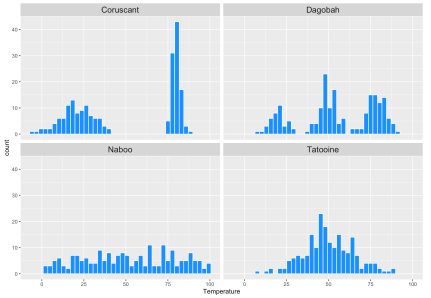


Q: What does the distribution of sample means look like?

Sampling Distributions are Approximately Normal

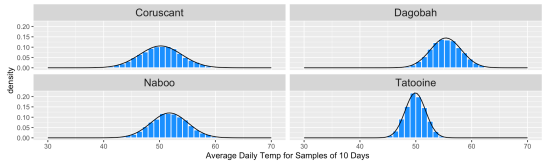
The **sampling distribution** for *each* planet appeared approximately Normal with $n = 50$, regardless of the shape of the population distribution.





Sampling Distributions are Approximately Normal

The **sampling distribution** for *each* planet appeared approximately Normal with $n = 50$, regardless of the shape of the population distribution.



We've seen this before! As n increases,

- Sampling distributions become more Normal
- Variance decreases
- Mean doesn't change; it is the parameter.

The Central Limit Theorem (CLT)

Theorem: Central Limit Theorem

Suppose a simple random sample of size n is drawn from a population with finite mean μ and finite standard deviation σ. Let $\bar{x}$ be the sample mean. When n is large, then approximately

$$\bar{x} \sim Normal\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$$

- A proof of the CLT requires more advanced techniques in probability (See Math 291).
- We have gained intuition already for the CLT by examining sampling distributions!
- We will use the CLT to conduct hypothesis tests/confidence intervals *without* simulation.

- **Even when the world is strange and chaotic at the individual level, averages follow clear, predictable rules that we can use to draw conclusions!**

The CLT is a BIG deal!

1. When conducting inference, our statistic is often a sample mean
   - A proportion is a sample mean of sorts too!

2. For nearly any population distribution, the sample mean is approximately Normal (when n is large enough)

3. The CLT gives a formula to calculate the mean and variance of a sample mean.
   - We can estimate these using our data
   - We can then quickly approximate what the sampling distribution looks like

Next time:

- We'll use the approximate distribution of the sample statistic (given by the CLT) to construct confidence intervals and to conduct hypothesis tests!